

CBSE 2027 Chemistry — Predicted Paper (Set 3 of 10)

Based on the official CBSE blueprint, historical PYQ trends, and the core conceptual areas repeatedly tested, here is Set 3 of the highly probable predicted board paper. All questions are distinct from previous sets and specifically target high-frequency numericals, reasoning, and named reactions.

SECTION A: Multiple Choice & Assertion-Reason Questions (1 Mark Each x 16 = 16 Marks)

- Q1.** Which of the following 0.1 M aqueous solutions will have the highest boiling point? [1] (a) Glucose (b) NaCl (c) $BaCl_2$ (d) Urea * **Chapter:** Solutions | **Confidence:** Very High
- Q2.** The unit of cell constant for a conductivity cell is: [2] (a) $\Omega^{-1}cm^{-1}$ (b) cm^{-1} (c) $\Omega \cdot cm$ (d) $S \cdot cm^2mol^{-1}$ * **Chapter:** Electrochemistry | **Confidence:** High
- Q3.** For a first-order reaction, a plot of half-life ($t_{1/2}$) versus initial concentration $[R]_0$ is: [3] (a) A straight line passing through the origin (b) A straight line parallel to the concentration axis (c) A parabola (d) A hyperbola * **Chapter:** Chemical Kinetics | **Confidence:** High
- Q4.** Which of the following elements is NOT regarded as a transition metal despite being in the d-block? [4] (a) Scandium (b) Copper (c) Zinc (d) Iron * **Chapter:** d- and f-Block Elements | **Confidence:** Very High
- Q5.** Which among the following is a hexadentate chelating ligand? [5] (a) Oxalate (b) Ethylenediamine (c) Ethane-1,2-diamine (d) EDTA * **Chapter:** Coordination Compounds | **Confidence:** High
- Q6.** Which of the following alkyl halides represents a chiral molecule? [6] (a) 1-Chlorobutane (b) 2-Chlorobutane (c) 1-Chloro-2-methylpropane (d) 2-Chloro-2-methylpropane * **Chapter:** Haloalkanes and Haloarenes | **Confidence:** Very High
- Q7.** Phenol, when treated with acidified sodium dichromate ($Na_2Cr_2O_7/H_2SO_4$), undergoes oxidation to give: [7, 8] (a) Benzoquinone (b) Salicylic acid (c) Benzene (d) Picric acid * **Chapter:** Alcohols, Phenols and Ethers | **Confidence:** High
- Q8.** The catalyst/reagent used in the Rosenmund reduction of an acid chloride to an aldehyde is: [9, 10] (a) $Zn(Hg)/HCl$ (b) $H_2, Pd/BaSO_4$ (c) $SnCl_2/HCl$ (d) $LiAlH_4$ * **Chapter:** Aldehydes, Ketones and Carboxylic Acids | **Confidence:** Very High
- Q9.** The Carbylamine reaction, producing an offensive smell of isocyanide, is an exclusive test for: [11] (a) Primary amines (both aliphatic and aromatic) (b) Secondary aliphatic amines (c) Tertiary aromatic amines (d) Quaternary ammonium salts * **Chapter:** Amines | **Confidence:** Very High
- Q10.** Deficiency of Vitamin C (Ascorbic acid) causes which of the following diseases? [12] (a) Night blindness (b) Scurvy (c) Rickets (d) Beri-beri * **Chapter:** Biomolecules | **Confidence:**

High

Q11. When chlorobenzene is heated with sodium metal in the presence of dry ether, biphenyl is formed. This reaction is known as the: [13] (a) Wurtz reaction (b) Fittig reaction (c) Wurtz-Fittig reaction (d) Sandmeyer reaction * **Chapter:** Haloalkanes and Haloarenes | **Confidence:** Very High

Q12. The oxidation state of Nickel (Ni) in the coordination complex $[Ni(CO)_4]$ is: [14] (a) +2 (b) +4 (c) 0 (d) -2 * **Chapter:** Coordination Compounds | **Confidence:** High

Directions for Q13 to Q16: Two statements are given—one labeled Assertion (A) and the other Reason (R). Select the correct answer from the codes (a), (b), (c), and (d). (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.

Q13. Assertion (A): Osmotic pressure is the preferred colligative property for determining the molar mass of macromolecules like proteins. **Reason (R):** Osmotic pressure values are measurable at room temperature and the property relies on molarity instead of molality. [1, 15] * **Chapter:** Solutions | **Confidence:** Very High

Q14. Assertion (A): The acid-catalyzed hydrolysis of an ester is a pseudo-first-order reaction. **Reason (R):** Water is present in such a large excess that its concentration remains practically constant throughout the reaction. [16] * **Chapter:** Chemical Kinetics | **Confidence:** High

Q15. Assertion (A): The actinoid contraction from element to element is greater than the lanthanoid contraction. **Reason (R):** The 5f electrons provide a poorer shielding effect from the increasing nuclear charge than the 4f electrons. [17, 18] * **Chapter:** d- and f-Block Elements | **Confidence:** Very High

Q16. Assertion (A): o-Nitrophenol is steam volatile whereas p-nitrophenol is not. **Reason (R):** o-Nitrophenol exhibits intramolecular hydrogen bonding, preventing molecular association. [7] * **Chapter:** Alcohols, Phenols and Ethers | **Confidence:** High

SECTION B: Very Short Answer Questions (2 Marks Each x 5 = 10 Marks)

Q17. State Raoult's law for a solution containing volatile liquids. How is Raoult's law considered a special case of Henry's Law? [19] * **Chapter:** Solutions | **Confidence:** High

Q18. Draw the *cis* and *trans* geometrical isomers of $[Pt(NH_3)_2Cl_2]$. Which of these two isomers possesses a zero dipole moment? [20] * **Chapter:** Coordination Compounds | **Confidence:** High

Q19. Explain why the conductivity of an electrolytic solution decreases with dilution, whereas its molar conductivity increases. [21, 22] * **Chapter:** Electrochemistry | **Confidence:** Very High

Q20. Give one simple chemical test to distinguish between the following pairs of amines: (i) Aniline and N-methylaniline. [23] (ii) Ethylamine and Ammonia. [24] * **Chapter:** Amines | **Confidence:** Very High

Q21. Define Activation Energy (E_a). How does the addition of a positive catalyst affect the activation energy and the rate of a chemical reaction? [25, 26] * **Chapter:** Chemical Kinetics | **Confidence:** High

SECTION C: Short Answer Questions (3 Marks Each x 7 = 21 Marks)

Q22. What chemical evidence happens when D-glucose is treated with the following reagents? Write the corresponding equations. (i) Hydrogen Iodide (HI) [27] (ii) Nitric acid (HNO_3) [28] (iii) Hydrogen cyanide (HCN) [28] * **Chapter:** Biomolecules | **Confidence:** Very High

Q23. Write the balanced chemical equations for the following essential conversions: (i) Cumene to Phenol [29] (ii) Ethanol to Ethene (using concentrated H_2SO_4 at 443 K) [30] (iii) Phenol to Picric Acid (2,4,6-trinitrophenol) [8] * **Chapter:** Alcohols, Phenols and Ethers | **Confidence:** High

Q24. Outline the synthetic steps to bring about the following organic conversions: (i) Propene to 1-propanol [31] (ii) 1-Chlorobutane to n-octane [31] (iii) Benzene to biphenyl [31] * **Chapter:** Haloalkanes and Haloarenes | **Confidence:** High

Q25. Give simple, distinct chemical tests to distinguish between the following pairs of carbonyl compounds: (i) Pentan-2-one and Pentan-3-one [32] (ii) Benzaldehyde and Acetophenone [32] (iii) Ethanal and Propanal [32] * **Chapter:** Aldehydes, Ketones and Carboxylic Acids | **Confidence:** Very High

Q26. Account for the following inorganic observations: (i) Transition metals and their compounds show highly effective catalytic activity. [33] (ii) Zinc, Cadmium, and Mercury are relatively soft metals. [34] (iii) Cr^{2+} is a strong reducing agent in aqueous solution. [35, 36] * **Chapter:** d- and f-Block Elements | **Confidence:** Very High

Q27. What are non-ideal solutions? Explain the characteristics of a non-ideal solution exhibiting a *positive deviation* from Raoult's law with the help of a suitable chemical example (e.g., ethanol and acetone) and state the expected signs for ΔH_{mix} and ΔV_{mix} . [37, 38] * **Chapter:** Solutions | **Confidence:** High

Q28. Calculate the mass of copper deposited when a steady current of 2.0 amperes is passed for 30 minutes through an aqueous solution of copper sulphate ($CuSO_4$). (Given: Molar mass of Cu = 63.5 g mol^{-1} , $1F = 96500 \text{ C mol}^{-1}$). [39] * **Chapter:** Electrochemistry | **Confidence:** Very High

SECTION D: Case Study-Based Questions (4 Marks Each x 2 = 8 Marks)

Q29. Case Study 1: Biomolecules (Protein Structure and Denaturation) Proteins are high-molecular-mass polymers of alpha-amino acids connected by a specific type of amide linkage. They exhibit highly complex structural organizations—primary, secondary, tertiary, and

quaternary. When a protein in its native, functional form is subjected to drastic physical changes (such as high temperature) or chemical changes (like a shift in pH), the hydrogen bonds are disturbed. Due to this, globules unfold and helices uncoil, leading the protein to lose its biological activity in a process termed denaturation. [40, 41] **(i)** What is the specific name of the amide linkage that forms the bond between constituent amino acids in proteins? (1 Mark) [42] **(ii)** Name the two most common macroscopic shapes of the secondary structure found in proteins. (1 Mark) [12] **(iii)** During the denaturation of a protein, which structural levels are disrupted and which remain intact? Explain the concept of a Zwitterion with respect to amino acids. (2 Marks) [41, 43] * **Chapter:** Biomolecules | **Confidence:** High

Q30. Case Study 2: Coordination Compounds (Crystal Field Theory) Crystal Field Theory (CFT) considers the metal-ligand coordinate bond to be a purely ionic interaction arising from electrostatic forces. In an octahedral complex, the approach of six negatively charged ligands causes the five degenerate d-orbitals of the central metal ion to split into two separate sets of different energies: a lower energy set (t_{2g}) and a higher energy set (e_g). The energy difference between these two sets is the crystal field splitting energy (Δ_o). This splitting phenomenon is responsible for the distinct color and magnetic properties of various coordination compounds. [44, 45] **(i)** Which set of d-orbitals (t_{2g} or e_g) experiences greater repulsion and is raised to a higher energy level in an octahedral crystal field? (1 Mark) [45] **(ii)** What is meant by the spectrochemical series? (1 Mark) [46] **(iii)** Based on electron transitions, explain why the complex $[Ti(H_2O)_6]^{3+}$ is colored (violet) in aqueous solution whereas the complex $[Sc(H_2O)_6]^{3+}$ is completely colorless. (2 Marks) [18, 36] * **Chapter:** Coordination Compounds | **Confidence:** Very High

SECTION E: Long Answer Questions (5 Marks Each x 3 = 15 Marks)

Q31. (A) Chemical Kinetics: Rate Equations & Numericals (i) Derive the integrated rate equation for a first-order chemical reaction. [47] (3 Marks) (ii) A first-order reaction possesses a specific rate constant of $1.15 \times 10^{-3} \text{ s}^{-1}$. How long will 5 g of this reactant take to reduce to exactly 3 g? (Given: $\log 5 = 0.699$, $\log 3 = 0.477$). [48] (2 Marks) * **OR (B)** (i) Define a pseudo-first-order reaction with the help of a suitable chemical example (e.g., ester hydrolysis). [16] (2 Marks) (ii) The half-life for the radioactive decay of Carbon-14 (^{14}C) is 5730 years. An archaeological artifact containing wood was found to have only 80% of the ^{14}C activity found in a living tree. Estimate the age of the wooden sample. (Given: $\log 1.25 = 0.0969$). [49] (3 Marks) * **Chapter:** Chemical Kinetics | **Confidence:** Very High

Q32. (A) Aldehydes, Ketones and Carboxylic Acids An organic compound **(A)** with the molecular formula $\text{C}_3\text{H}_6\text{O}$ is highly resistant to oxidation but forms a compound **(B)** with the formula $\text{C}_3\text{H}_8\text{O}$ upon catalytic reduction. Compound **(B)** reacts with HBr to form an alkyl bromide **(C)**. Compound **(C)**, upon treatment with alcoholic KOH, undergoes elimination to form an alkene **(D)** with the formula C_3H_6 . [50] (i) Deduce the structures and IUPAC names of compounds **A**, **B**, **C**, and **D**. (4 Marks) [50] (ii) Write the chemical equation specifically for the conversion of **(B)** to **(C)**. (1 Mark) [51] * **OR (B)** Complete the following important name

reactions by providing their balanced chemical equations: [9, 10] (i) Rosenmund reduction (ii) Stephen reaction (iii) Etard reaction (iv) Gattermann-Koch reaction (v) Hell-Volhard-Zelinsky (HVZ) reaction * **Chapter:** Aldehydes, Ketones and Carboxylic Acids | **Confidence:** Very High

Q33. (A) Electrochemistry (i) State Michael Faraday's First Law of Electrolysis. [52] (1 Mark) (ii) What is the function of a salt bridge in a standard Galvanic cell? [21] (1 Mark) (iii) Calculate the Electromotive Force (EMF) of the following electrochemical cell at 298 K:

$Fe_{(s)}|Fe^{2+}(0.001M)||H^+(1M)|H_{2(g)}(1\text{ bar})|Pt_{(s)}$ (Given: $E_{Fe^{2+}/Fe}^{\circ} = -0.44V$). [36, 53]

(3 Marks) * **OR (B)** (i) Define limiting molar conductivity (Λ_m°). Why does the molar conductivity of a weak electrolyte (like acetic acid) increase steeply upon high dilution? [22] (2 Marks) (ii) The conductivity of a 0.20 mol L^{-1} solution of KCl at 298 K is 0.0248 S cm^{-1} .

Calculate its molar conductivity (Λ_m) and determine its degree of dissociation if the limiting molar conductivity (Λ_m°) for KCl is $149.9\text{ S cm}^2\text{ mol}^{-1}$. [54] (3 Marks) * **Chapter:**

Electrochemistry | **Confidence:** Very High

CBSE Class 12 (2027) — AI-Predicted via PYQ/Blueprint Analysis. Probability-weighted guidance, not actual exam questions.